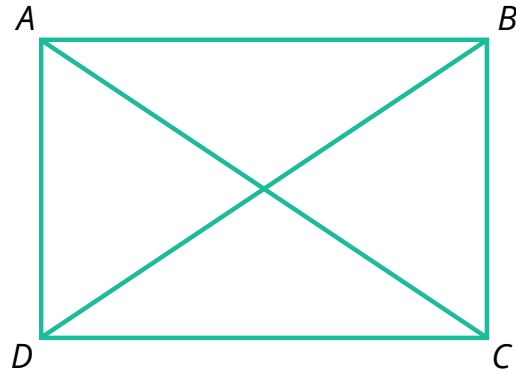


Properties of rectangles, rhombuses, squares

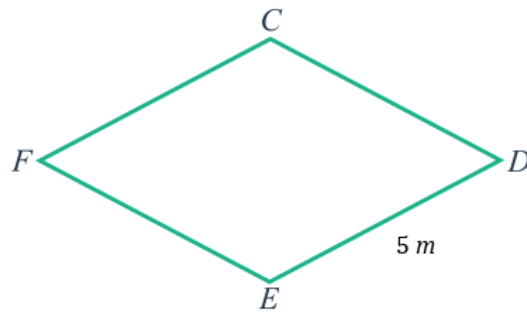
- 1 Consider the given rectangle $ABCD$.
Suppose that $\overline{AC} = 16$ m. Find:

a $\overline{BD}$ b $m\angle BCD$

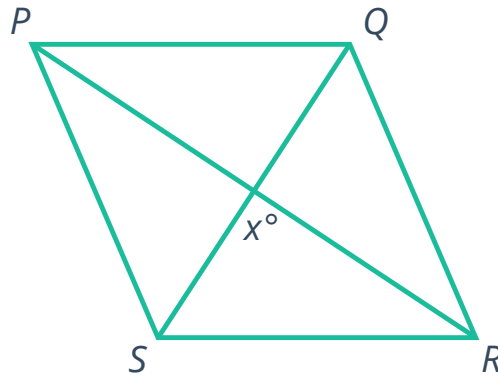


- 2 Consider the given rhombus $CDEF$.
Find:

a $\overline{CD}$ b $\overline{EF}$ c $\overline{FC}$

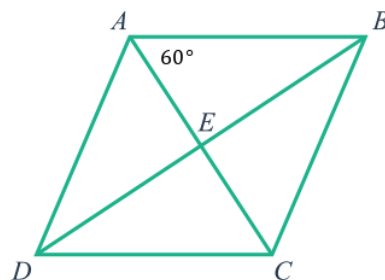


- 3 Consider the given rhombus $PQRS$.
Find the value of x .



- 4 Consider the given rhombus $ABCD$.
Find:

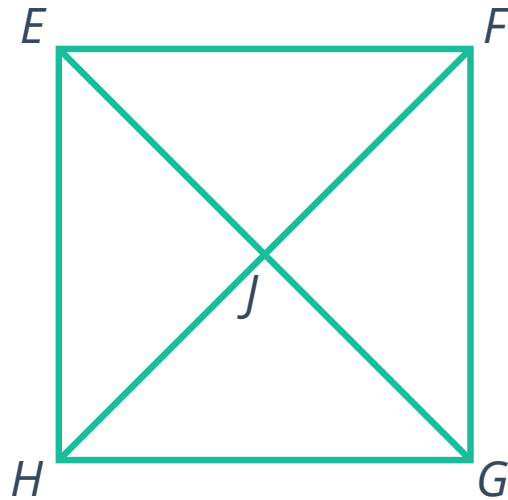
a $m\angle BEC$ b $m\angle BCE$
c $m\angle ECD$ d $m\angle BCD$





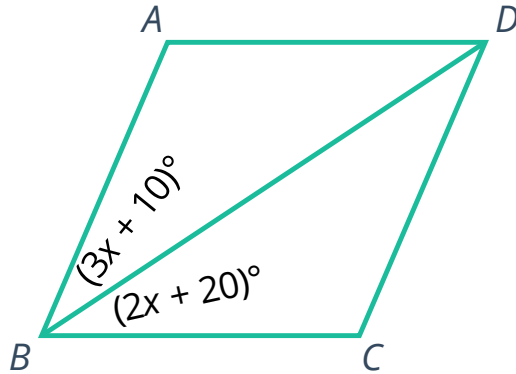
- 5 Consider the given square $EFGH$.
Suppose that $JG = 19$ cm. Find:

- a $\overline{EJ}$ b $\overline{FH}$
c $m\angle EJF$ d $m\angle JFG$



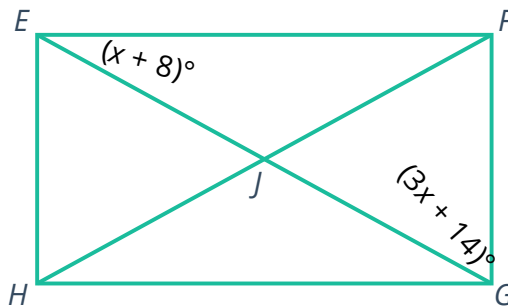
- 6 Consider the given rhombus $ABCD$.

- a Solve for x . b Find $m\angle ABD$.



- 7 Consider the given rectangle $EFGH$.

- a Solve for x . b Find $m\angle HJG$.

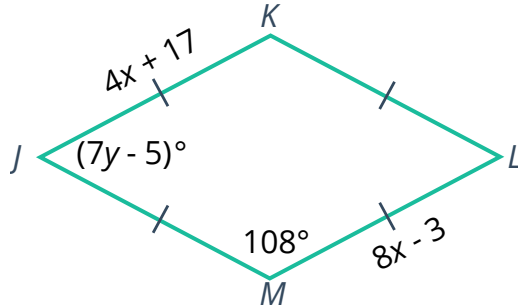


8 For each of the following parallelograms:

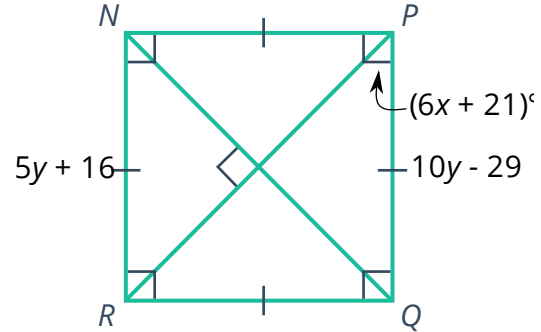


- i Classify the parallelogram as rectangle, rhombus, or square.
- ii Solve for x
- iii Solve for y

a



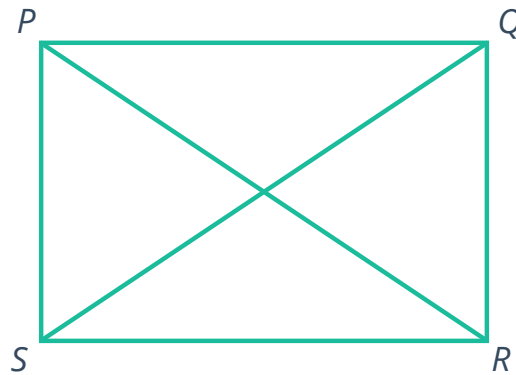
b



9 In the given rectangle $PQRS$:

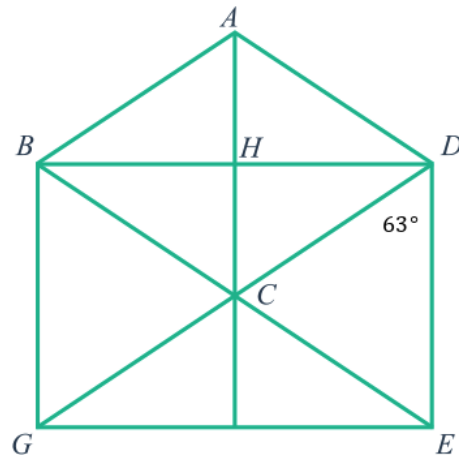
$$\begin{aligned}\overline{PQ} &= x + 50; \\ \overline{PR} &= 2x + 10; \text{ and} \\ \overline{SQ} &= 4x - 130\end{aligned}$$

- a Solve for x .
- b Find the length of $\overline{PQ}$.

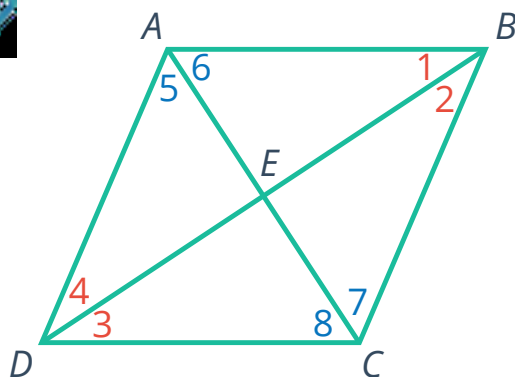


10 In the given diagram, $BDEG$ is a rectangle and $ABCD$ is a rhombus.
Find:

- | | |
|-----------------|-----------------|
| a $m\angle BDE$ | b $m\angle BDC$ |
| c $m\angle ADC$ | d $m\angle BAD$ |
| e $m\angle CAD$ | f $m\angle ACD$ |



- 11 Consider the following proof based on the given rhombus, $ABCD$:



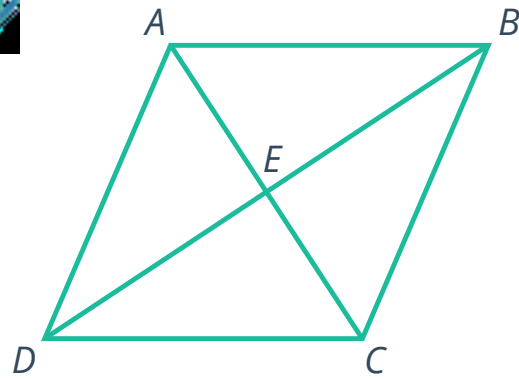
To prove: $\angle 1 \cong \angle 2 \cong \angle 3 \cong \angle 4 \cong \angle 5 \cong \angle 6 \cong \angle 7 \cong \angle 8$

Statements	Reasons
1. $ABCD$ is a rhombus	Given
2. $ABCD$ is a parallelogram $\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$	Definition of a rhombus
3. $\overline{AE} \cong \overline{EC}$ and $\overline{DE} \cong \overline{EB}$	If a quadrilateral is a parallelogram, then its diagonals bisect each other.
4. $\overline{AE} \cong \overline{AE}$, $\overline{BE} \cong \overline{BE}$, $\overline{CE} \cong \overline{CE}$, and $\overline{DE} \cong \overline{DE}$	Reflexive property of congruence
5. $\triangle AEB \cong \triangle CEB \cong \triangle CED \cong \triangle AED$	Side-side-side congruence theorem
6. $\angle 1 \cong \angle 2 \cong \angle 3 \cong \angle 4 \cong \angle 5 \cong \angle 6 \cong \angle 7 \cong \angle 8$	Corresponding parts of congruent triangles are congruent (CPCTC).

Determine whether each statement is true or false.

- a If a quadrilateral is a rhombus, then its diagonals bisect the angles at the vertices.
- b If a quadrilateral is a rhombus, then its diagonals are congruent.
- c If a quadrilateral has diagonals which bisect the angles at the vertices, then it is a rhombus.
- d If a quadrilateral is a parallelogram, then its diagonals bisect the angles at the vertices.

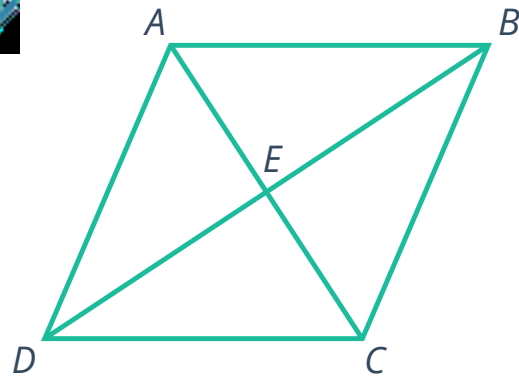
- 12 Complete the following proof to show that the given rhombus $ABCD$ is composed of four congruent triangles:



To prove: $ABCD$ is composed of four congruent triangles

Statements	Reasons
1. $ABCD$ is a rhombus	Given
2. $ABCD$ is a parallelogram $\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$	Definition of a rhombus
3. □	□
4. $\overline{AE} \cong \overline{AE}, \overline{BE} \cong \overline{BE},$ $\overline{CE} \cong \overline{CE}, \text{ and } \overline{DE} \cong \overline{DE}$	Reflexive property of congruence
5. $\triangle AEB \cong \triangle CEB \cong \triangle CED \cong \triangle AED$	Side-side-side congruence theorem

- 13 Consider the following proof based on the given rhombus, $ABCD$:



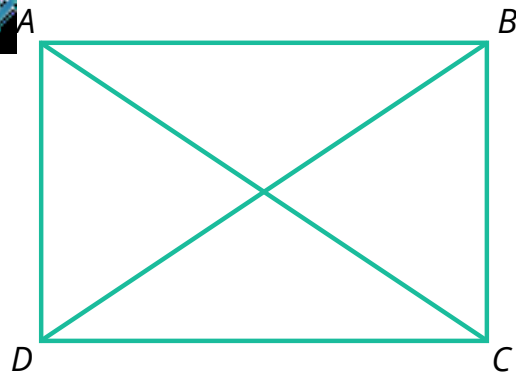
To prove: $\overline{AC} \perp \overline{BD}$

Statements	Reasons
1. $ABCD$ is a rhombus	Given
2. $\triangle AEB \cong \triangle CEB \cong \triangle CED \cong \triangle AED$	If a quadrilateral is a rhombus, then it is composed of four congruent triangles.
3. $\angle AEB \cong \angle AED$ (and two other angles in the center)	Corresponding parts of congruent triangles are congruent (CPCTC).
4. $\angle AEB$ and $\angle AED$ form a linear pair (and two other angles in the center)	Given
5. $\angle AEB$ and $\angle AED$ are right angles (and two other angles in the center)	Two congruent angles that form a linear pair are right angles.
6. $\overline{AC} \perp \overline{BD}$	Definition of perpendicular

Determine whether each statement is true or false.

- a If a quadrilateral is a rhombus, then its diagonals are perpendicular.
- b If a quadrilateral is a parallelogram, then its diagonals bisect the angles at the vertices.
- c If a quadrilateral is a rhombus, then its diagonals are congruent.
- d If a quadrilateral has diagonals that are perpendicular, then it is a rhombus.

- 14 Complete the following proof to show that in the given rectangle $ABCD$, $\overline{AC} \cong \overline{BD}$:



To prove: $\overline{AC} \cong \overline{BD}$

Statements	Reasons
1. $ABCD$ is a rectangle	Given
2. $ABCD$ is a parallelogram $\angle ADC$ and $\angle BCD$ are right angles	Definition of a rectangle
3. $\angle ADC \cong \angle BCD$	All right angles are congruent.
4. $\overline{AD} \cong \overline{BC}$	If a quadrilateral is a parallelogram, then its opposite sides are congruent.
5. □	□
6. $\triangle ACD \cong \triangle BDC$	Side-angle-side congruence theorem
7. $\overline{AC} \cong \overline{BD}$	Corresponding parts of congruent triangles are congruent (CPCTC).

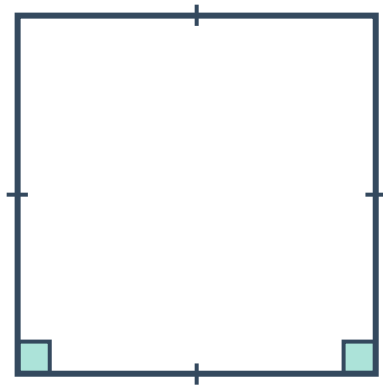
Additional questions

- 15 Determine whether each statement is true or false:
- a All quadrilaterals are parallelograms
 - b All rectangles are parallelograms
 - c All parallelograms are rhombi
 - d A parallelogram with four congruent sides is a square
 - e A rectangle with consecutive sides congruent is a square
 - f A rhombus is a parallelogram

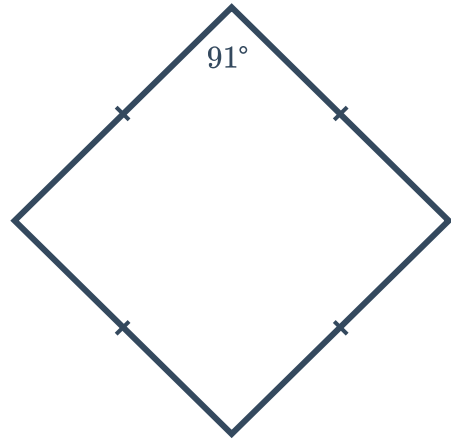
16 Classify each quadrilateral. Explain your reasoning.



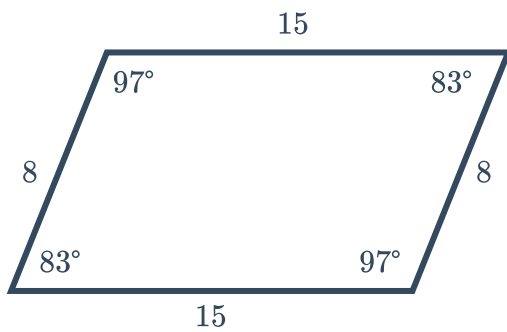
a



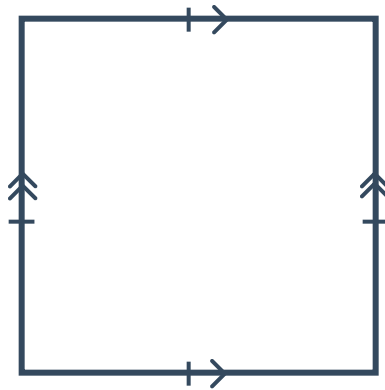
b



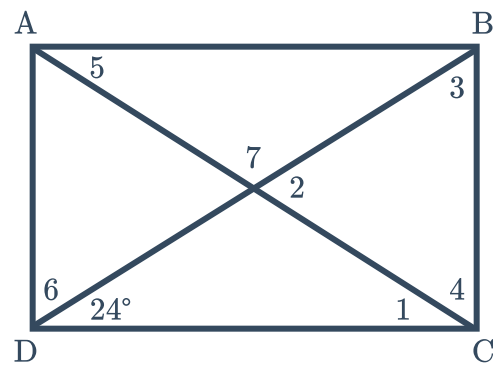
c



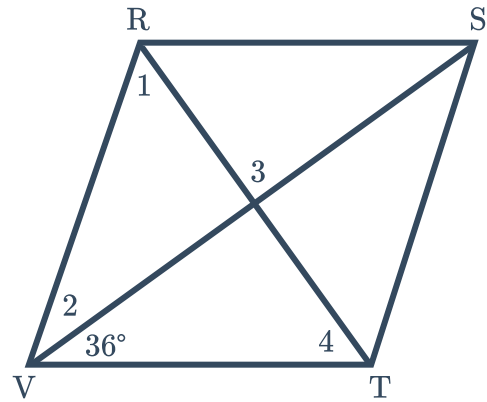
d



17 Given: $ABCD$ is a rectangle
Solve for each missing angle in the diagram.

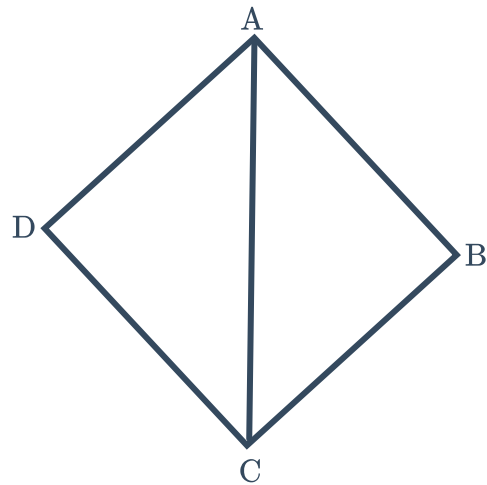


- 18 Given: $RSTV$ is a rhombus
Solve for each missing angle in the diagram.



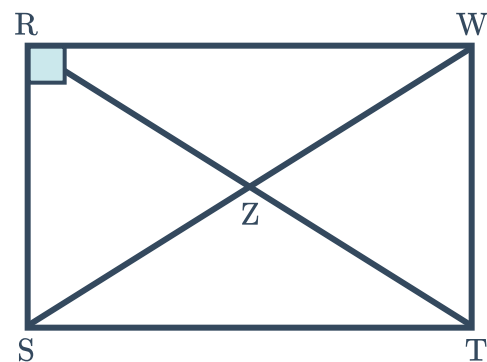
- 19 Given: $ABCD$ is a rhombus
 $m\angle DAC = 4x + 9$
 $m\angle DAB = 11x - 3$

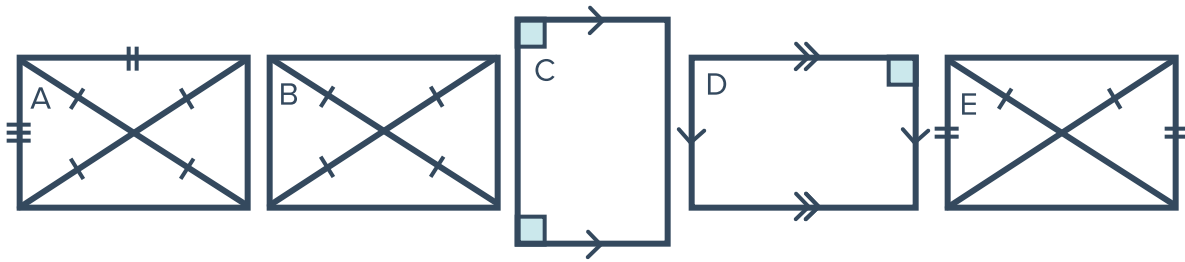
- Solve for x
- Solve for $m\angle BAC$
- Solve for $m\angle ABC$



- 20 $RZ = 4x + 10$
 $SW = 5x - 20$

- Classify the quadrilateral.
- Solve for x .

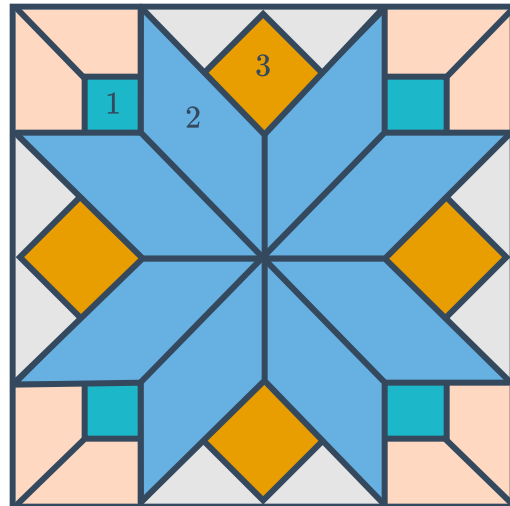




- a Based on the given markings, which of the following quadrilaterals is not a rectangle? Explain how you know.
- b What additional information could make the quadrilateral you found in part (b) a rectangle?

22 Stained glass windows are made by putting together various shapes, like those in the design below:
For quadrilaterals 1-3 in the design:

- a Make your best guess as to the classification of the quadrilateral.
- b Determine whether you could correctly classify the quadrilaterals using only a ruler.



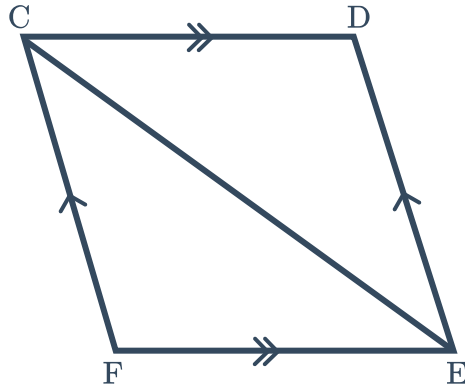
23 Sketch each of the following, labeling all sides and angles accordingly:

- a A parallelogram that is neither a rectangle or a rhombus.
- b A parallelogram with a pair of consecutive congruent sides and a pair of acute angles.
- c A parallelogram with a right angle.

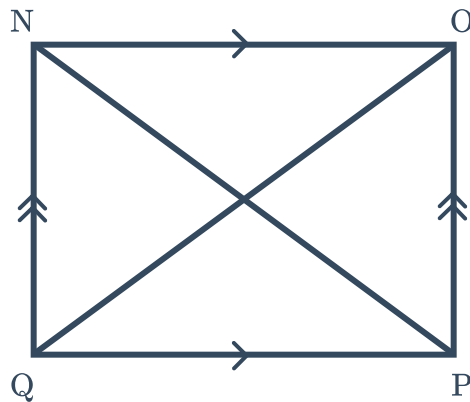
24 Determine whether the given information is enough to prove the given statement. Justify your answer.



- a Given: Parallelogram $CDEF$
Conclusion: $CDEF$ is a rhombus.



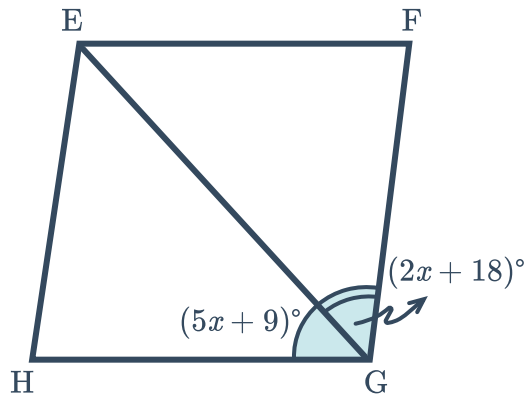
- b Given: Parallelogram $NOPQ$
 $\overline{NO} \cong \overline{PQ}$
Conclusion: $NOPQ$ is a rectangle.



c Given: Parallelogram $EFGH$

$$x = 3$$

Conclusion: $EFGH$ is a rhombus.



25 If $\overline{AB} \cong \overline{CD}$, show that $ABCD$ is not a rhombus.

